The Trichotomy of Property Testing Regular Languages

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LaBRI, Bordeaux MPI-SWS Kaiserslautern

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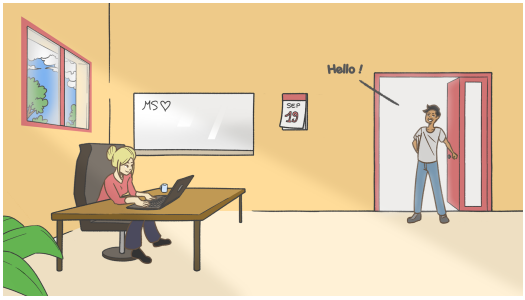
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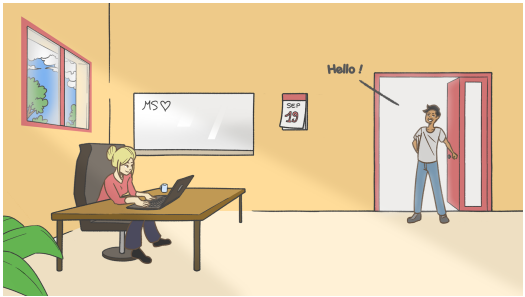
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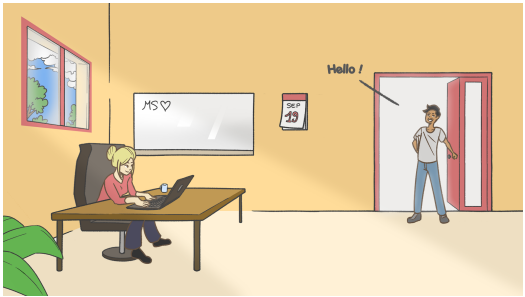
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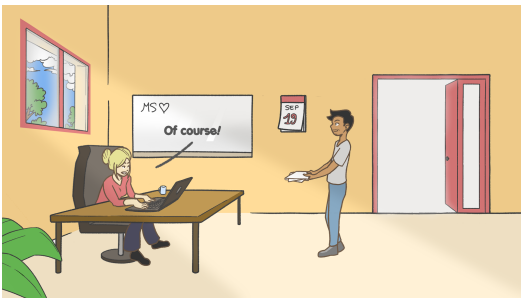
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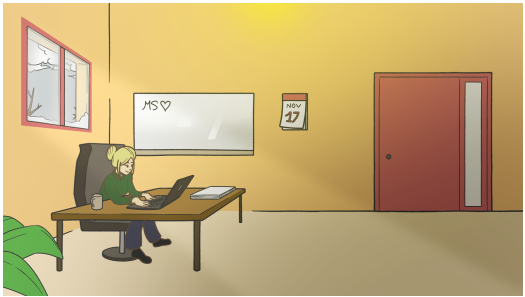
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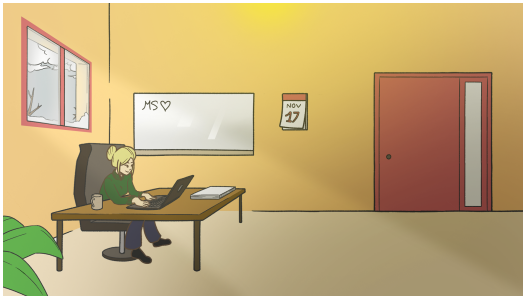
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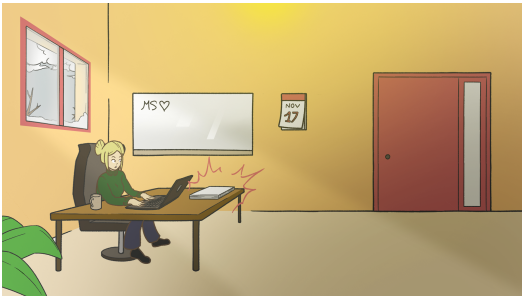
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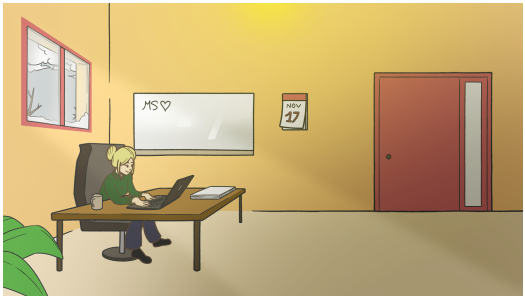
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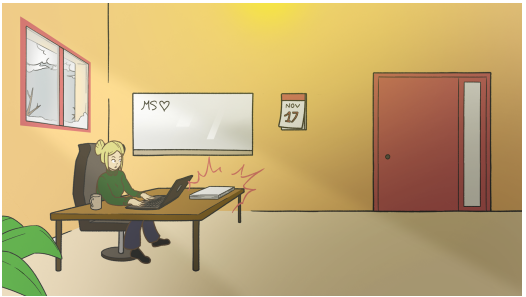
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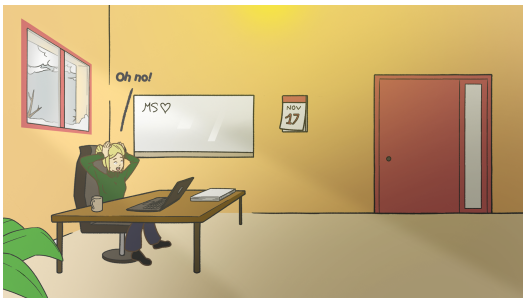
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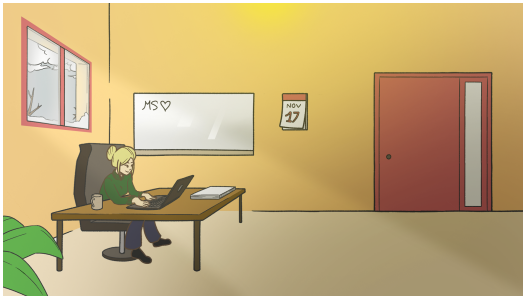
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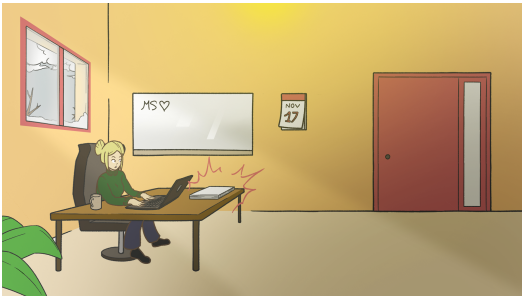
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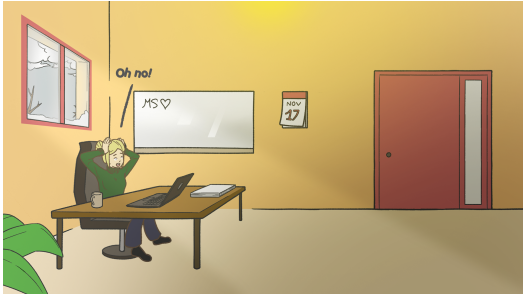
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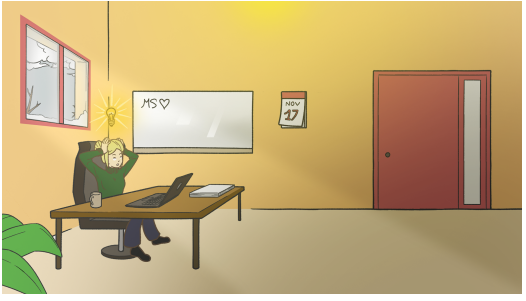
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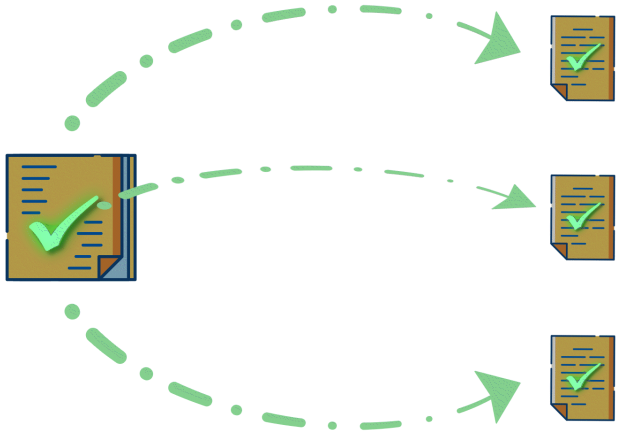

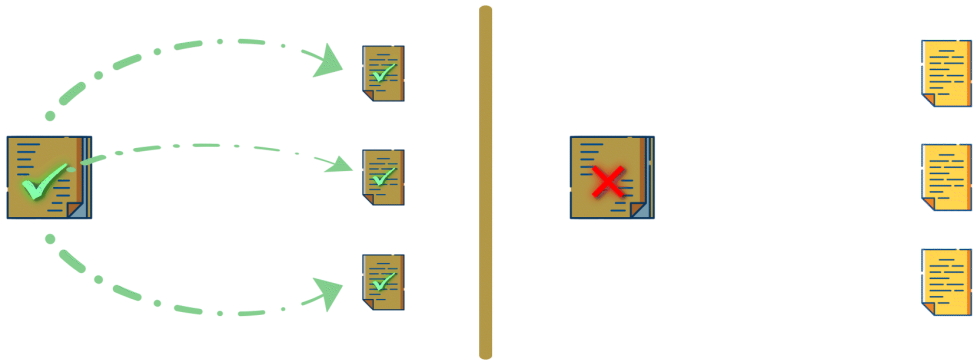

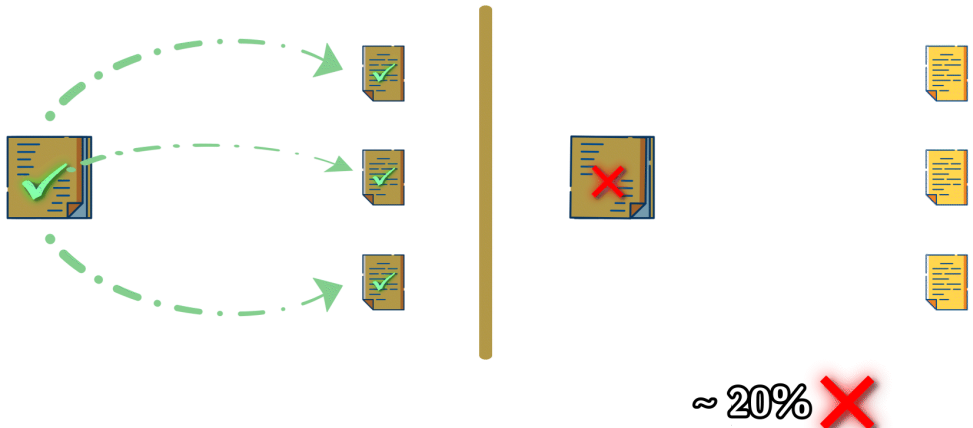
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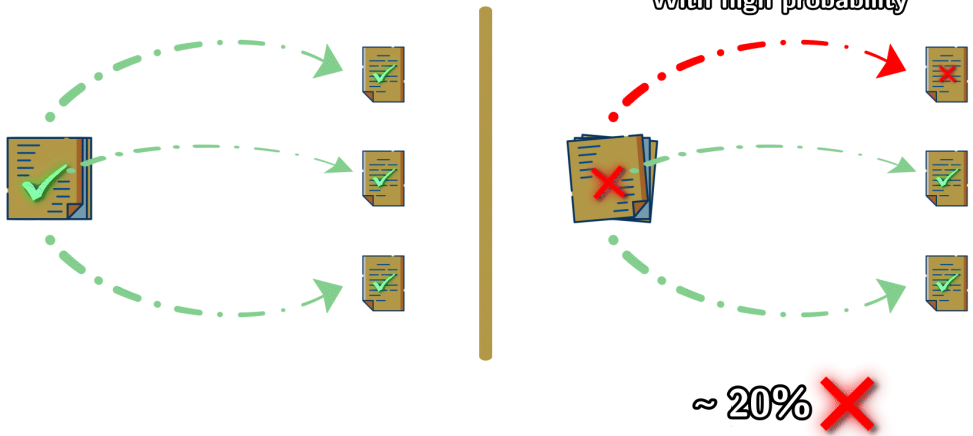
With high probability
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Property testing

Class of objects $\mathcal{C}$

Ex: words, trees, graphs, lists, matrices

Size function $|.|$

Ex: number of vertices+edges on graphs

Distance function d

Ex: edge-edit distance on graphs

Subclass P

Ex: connected graphs, k-colourable graphs

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Fix a parameter $\varepsilon > 0$.

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Fix a parameter $\varepsilon > 0$.

Property testing

Input: $obj \in \mathcal{C}$

Output: If $obj \in P \rightarrow$ **Yes**

If $d(obj, P) > \varepsilon |obj| \rightarrow$ **No** with probability $\geq 1/3$

Example

Words over $\{a, b\}$, equipped with the *Hamming distance*.

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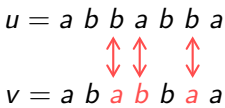
$$d(u,v) = \begin{cases} |\{i \mid u[i] \neq v[i]\}| \text{ if } |u| = |v| \\ +\infty \text{ if } |u| \neq |v| \end{cases}$$



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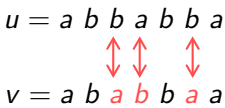

Property $P = a^*$

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Find an algorithm such that, on a word w,
If $w \in a^* \rightarrow$ **Yes**
If $d(w, a^*) > \varepsilon|w| \rightarrow$ **No** with constant positive probability

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independent of $|w|$

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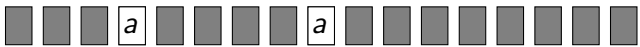

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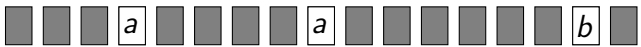

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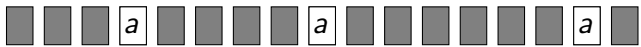

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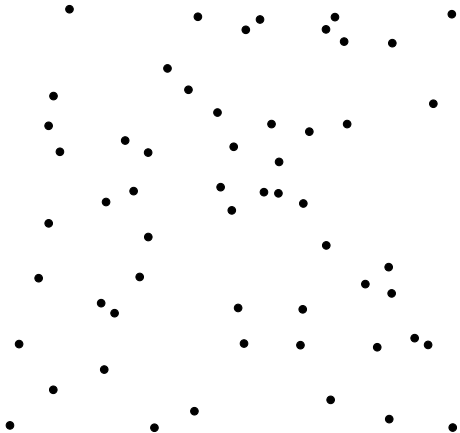
Example: connectivity



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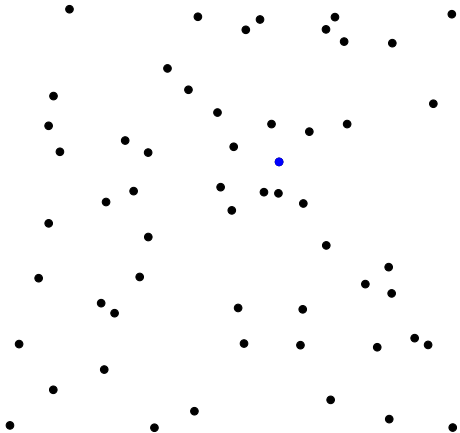
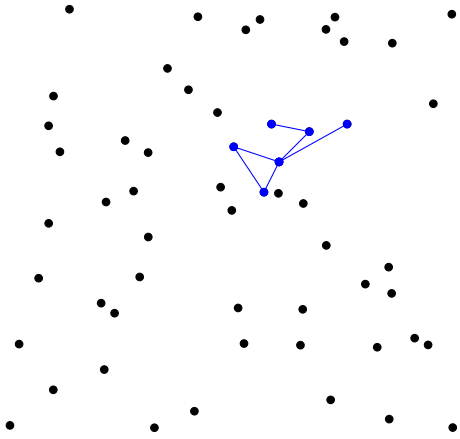


- Pick a random vertex

Example: connectivity



- Pick a random vertex
- Explore its connected component by DFS

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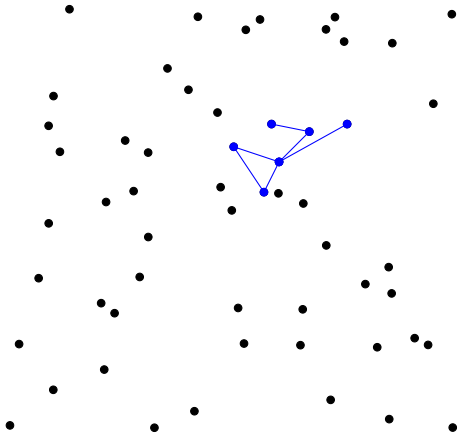

- Pick a random vertex
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- If it has size $< 2/\varepsilon$ then return No

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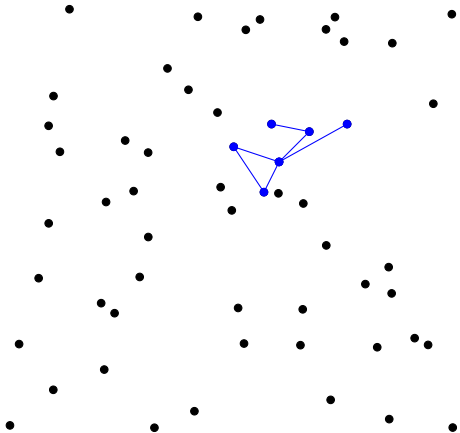
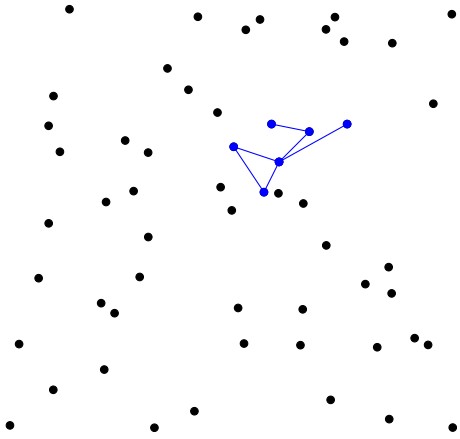


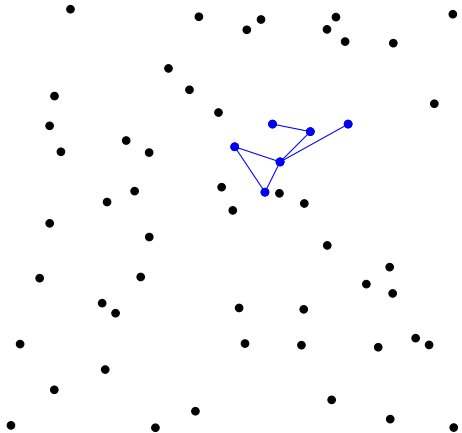
- Pick a random vertex
- Explore its connected component by DFS
- If it has size $< 2/\varepsilon$ then return No
- Otherwise pick another point etc...
- Return Yes if $> 1/\varepsilon$ points have been picked

Example: connectivity



for $0 \leq t \leq 1/\varepsilon$ **do**
    $v \leftarrow rand(V)$
    if $|SCC(v)| < 2/\varepsilon$ **then**
        return No
return Yes

Example: connectivity



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- G is connected $\Rightarrow$ Yes.
- $d(G, P_{connected}) \geq \varepsilon|V|$
  $\Rightarrow$ G has $\geq \varepsilon|V|$ connected components
  $\Rightarrow$ G has $\geq \varepsilon|V|/2$ connected components of size $\leq 2/\varepsilon$
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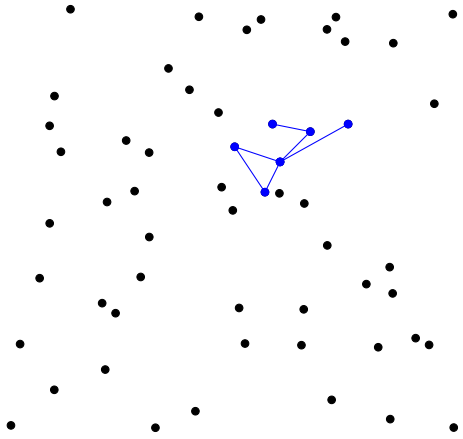

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$\mathcal{O}(1/\varepsilon^2)$ queries.

Back to words

Property testing on words

Input: $w \in \Sigma^*$

Output: If $w \in L \rightarrow$ **Yes**

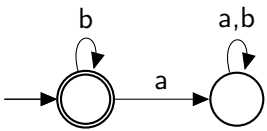
If $d(w, L) > \varepsilon|w| \rightarrow$ **No** with probability $\geq 1/3$

Problem

Find efficient property testing algorithms for regular languages.

Example

$L = b^*$ over $\Sigma = \{a, b\}$



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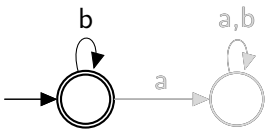

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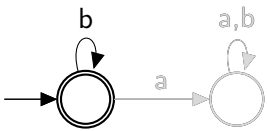

for $1 \leq k \leq 1/\varepsilon$ **do**
  $i \leftarrow rand(\{1, \ldots, |w|\})$
  if $w[i] = a$ **then**
    return No
return Yes

$\mathcal{O}(1/\varepsilon)$ queries

- $w \in L \Rightarrow$ **Yes**
- $d(w, L) > \varepsilon|w| \Rightarrow \mathbb{P}(\textbf{No}) \geq 1 - (1 - \varepsilon)^{1/\varepsilon} \geq 1/3$

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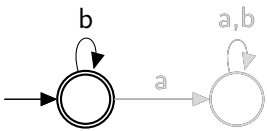

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Blocking factor = Word that is not a factor of any word in L.

$$\mathcal{BF}((a^+bb)^*a^*) = \Sigma^*(aba + bbb)\Sigma^*$$

[1] Alon, Krivelevich, Newman, Szegedy 2001

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$$\mathcal{BF}((a^+bb)^*a^*) = \Sigma^*(aba + bbb)\Sigma^*$$

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If L is recognised by a **strongly connected** automaton, then for all w,

$$d(w, L(\mathcal{A})) \geq \varepsilon|w| \Rightarrow w \text{ contains } \Omega(\varepsilon|w|) \text{ disjoint blocking factors.}$$

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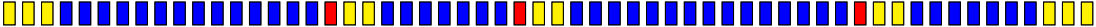
$w =$ 

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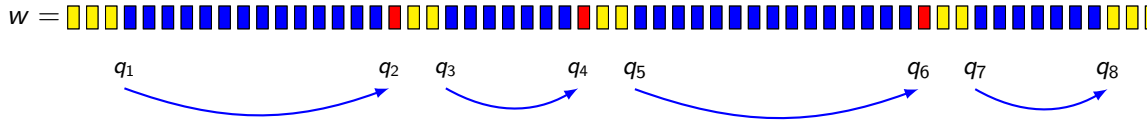

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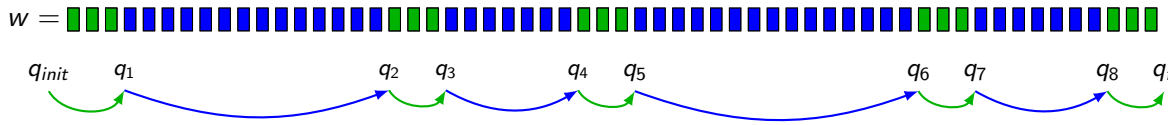

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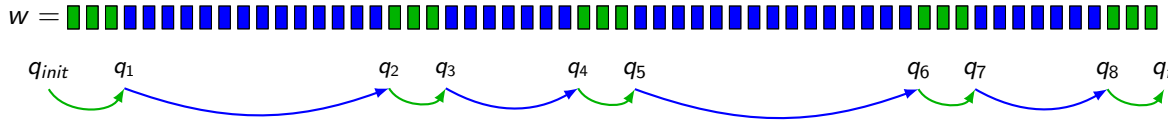


Theorem[1]

w has $\Omega(\varepsilon|w|)$ blocking factors $\Rightarrow$ w has $\Omega(\varepsilon|w|)$ blocking factors of size $\mathcal{O}(1/\varepsilon)$.

[1] Alon, Krivelevich, Newman, Szegedy 2001

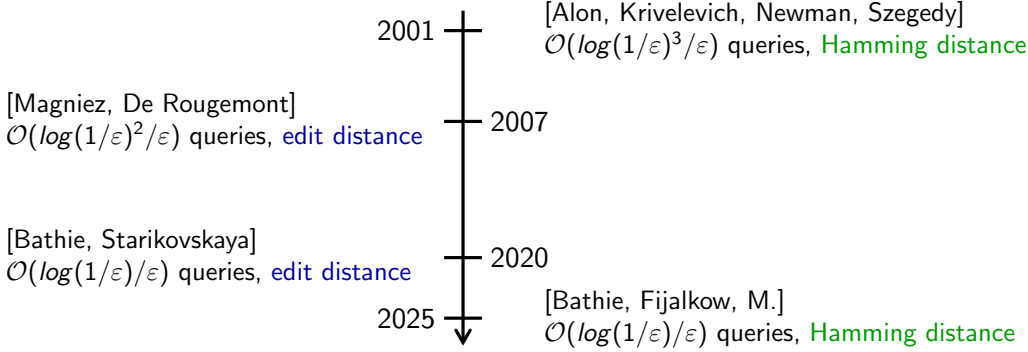
$d(w, L(\mathcal{A})) \geq \varepsilon|w| \Rightarrow w$ contains $M\varepsilon|w|$ blocking factors of size $\leq M/\varepsilon$. .

for $1 \leq k \leq M/\varepsilon$ **do**
  $i \leftarrow rand(\{1, \ldots, |w|\})$
  if $w[i - M/\varepsilon, i + M/\varepsilon] \in \mathcal{BF}(L)$ **then**
    return No
return Yes

Theorem

If L is recognised by a strongly connected automaton then it is testable with $\mathcal{O}(1/\varepsilon^2)$ queries.

Upper bounds



Theorem

Every regular language is testable using $\mathcal{O}(log(1/\varepsilon)/\varepsilon)$ queries under the Hamming distance.

for $0 \leq t \leq log(M/\varepsilon)$ **do**
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Sample $1/\varepsilon\ell$ factors of length 2ℓ

Lower bounds

Theorem[1]

The language $L = b^*$ over $\{a, b\}^*$ requires $\Omega(1/\varepsilon)$ queries.

[1] Alon, Krivelevich, Newman, Szegedy '01
[2] Bathie, Starikovskaya '21

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Key ingredient: *Yao's minimax principle*

- Choose a distribution $\mathcal{D}$ over inputs
- Show lower bound on expected complexity of deterministic algorithms over $\mathcal{D}$.

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Yao's minimax principle

Take an arbitrary distribution $\mathcal{D}$ over inputs, then for all randomised algorithm $\mathbf{R}$,

$$\min_{\mathbf{A}\ det.} \mathbb{E}_{in\sim\mathcal{D}}(cost(\mathbf{A}, in)) \leq \max_{in\in I} \mathbb{E}_{r\sim\mathcal{U}}(cost(\mathbf{R}, r, in))$$

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If $\mathbf{R}$ uses d queries, we infer

$$\min_{\substack{\mathbf{A}\ det.\\ d \text{ queries}}} \mathbb{P}_{in\sim\mathcal{D}}(\mathbf{A} \text{ incorrect on } in) \leq \max_{in\in I} \mathbb{P}_{r\sim\mathcal{U}}(\mathbf{R} \text{ incorrect on } in)$$

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Every property testing algorithm needs $> d$ queries.

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The language $L = b^*$ over $\{a, b\}^*$ requires $\Omega(1/\varepsilon)$ queries.

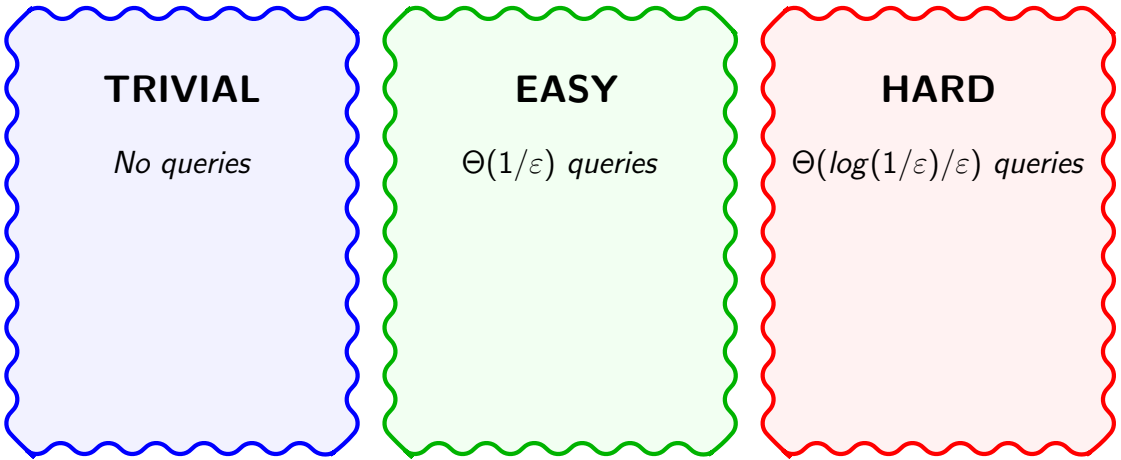
Choose $\mathcal{D}$ so that:

- b^n with proba $1/2$
- $b^{i\varepsilon/n}a^{\varepsilon/n}b^{n-(i+1)\varepsilon/n}$ with proba $\varepsilon/2$.

A deterministic property testing algorithm:

- must answer Yes if it only sees b
- has a probability $> 2/3$ to only see b if it queries $< \varepsilon/3$ letters.

Trichotomy Theorem for strongly connected automata



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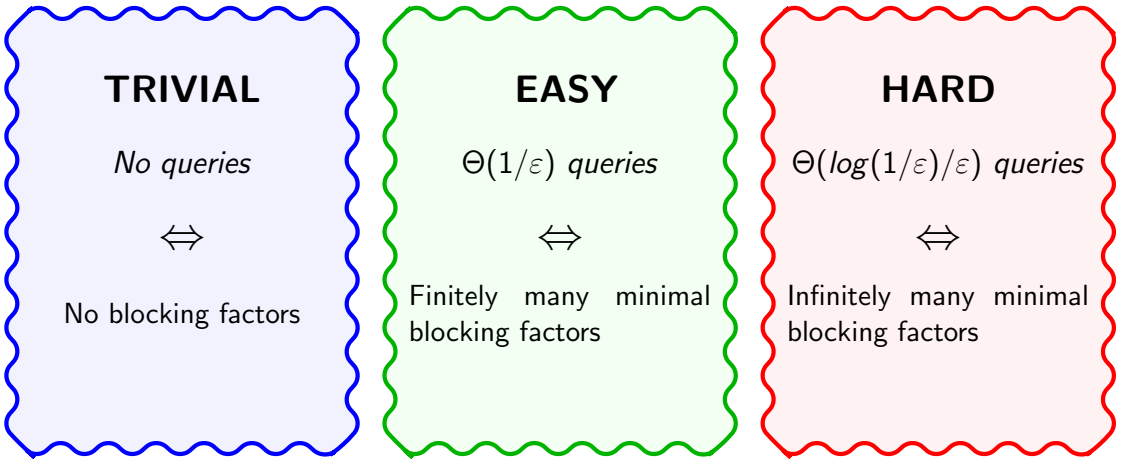

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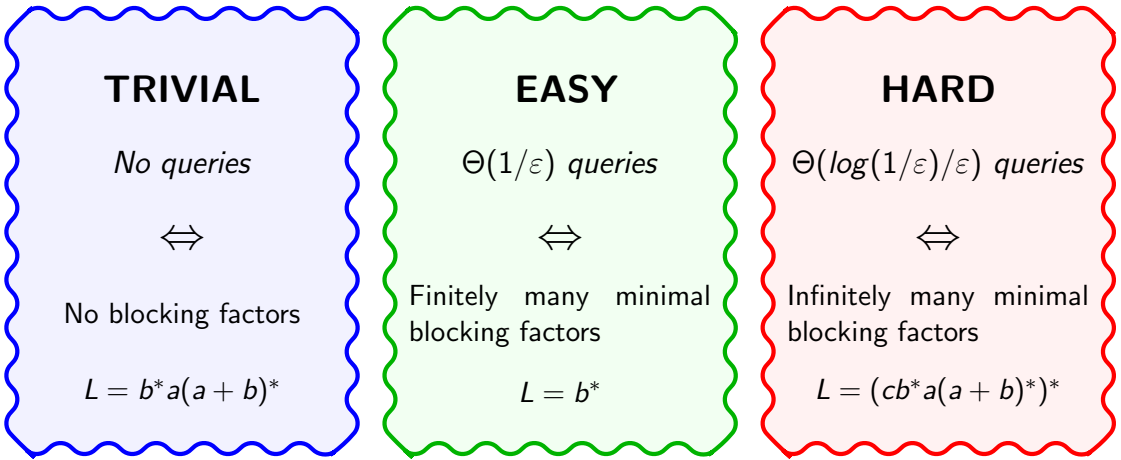


Trichotomy Theorem for strongly connected automata

TRIVIAL

No queries

$\Leftrightarrow$

No blocking factors

$L = b^*a(a+b)^*$

EASY

$\Theta(1/\varepsilon)$ *queries*

$\Leftrightarrow$

Finitely many minimal blocking factors

$L = b^*$

HARD

$\Theta(log(1/\varepsilon)/\varepsilon)$ *queries*

$\Leftrightarrow$

Infinitely many minimal blocking factors

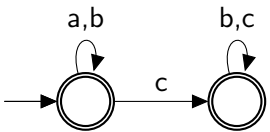
$L = (cb^*a(a+b)^*)^*$

What about automata with multiple SCCs?

Example

$L = (a + b)^*(b + c)^*$ over $\Sigma = \{a, b, c\}$.

Minimal blocking factors $= cb^*a$.



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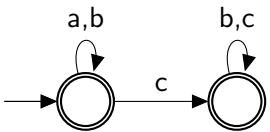

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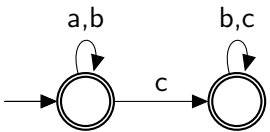

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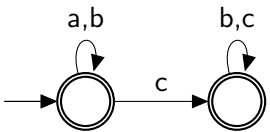

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Blocking sequence = Sequence of words that do not appear in that order in a word $w \in L$.

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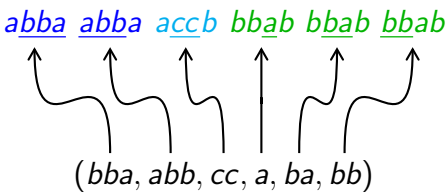
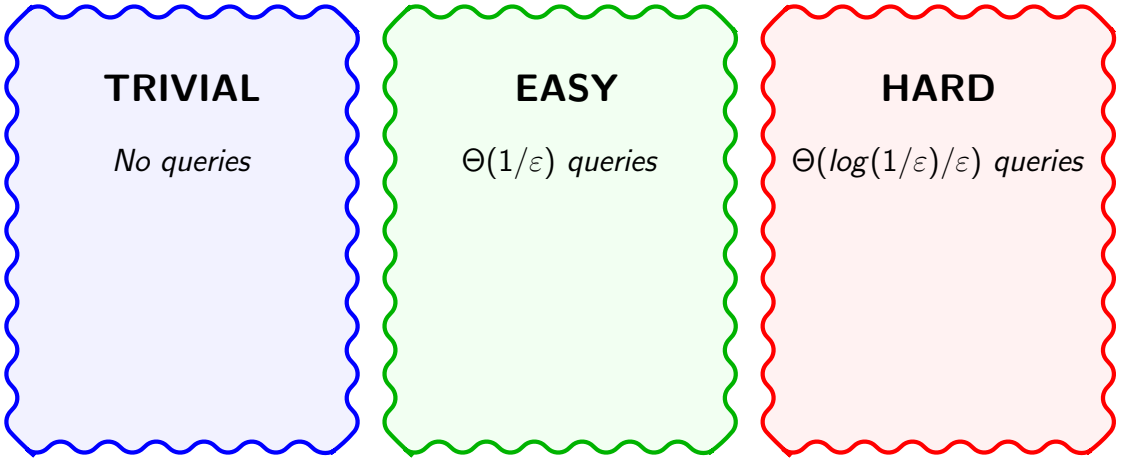
$(abba, accb, bbab)$



$\mathcal{MBS}(L)$ = minimal blocking sequences of L.

$$\mathcal{MBS}(a^* b^* c^*) = \{(b, a), (c, a), (c, b)\}$$

Trichotomy Theorem



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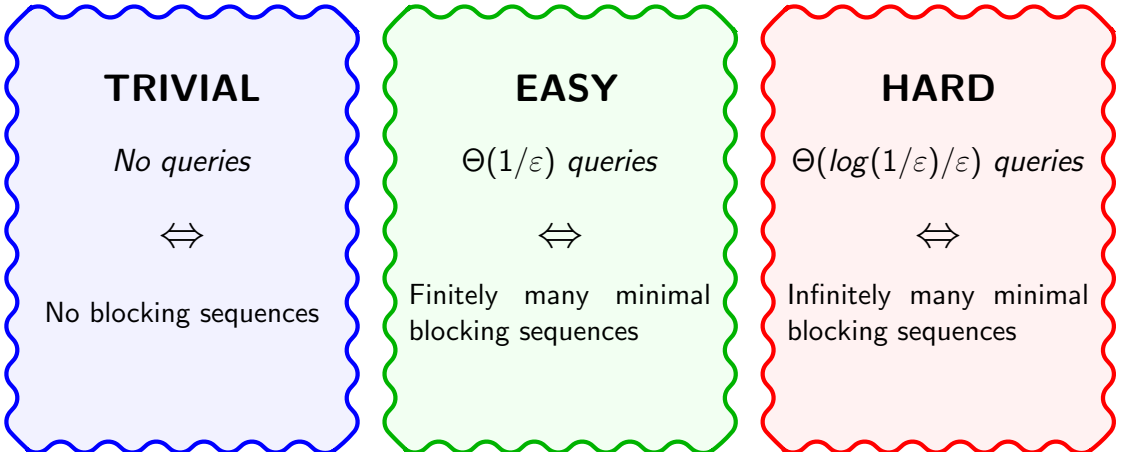

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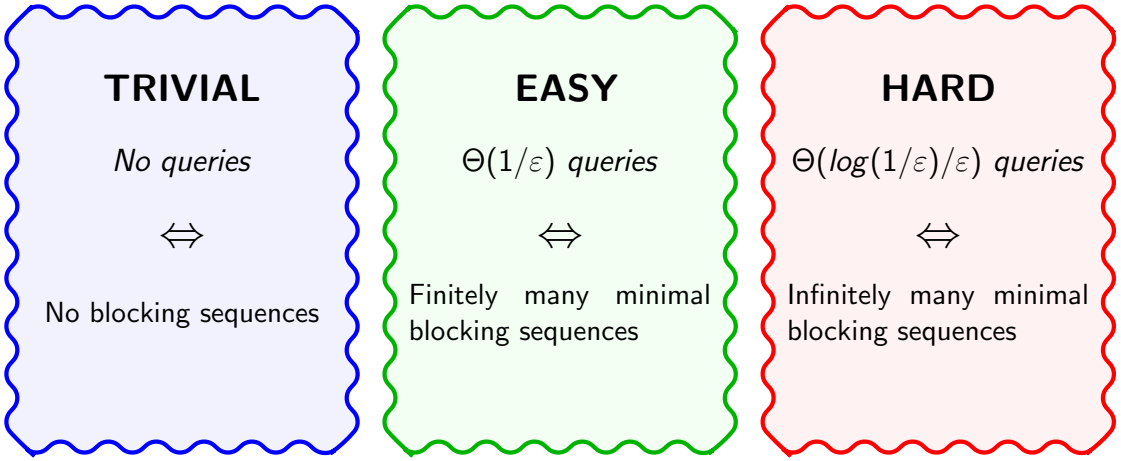

Lower bounds obtained by reduction to the strongly connected case.

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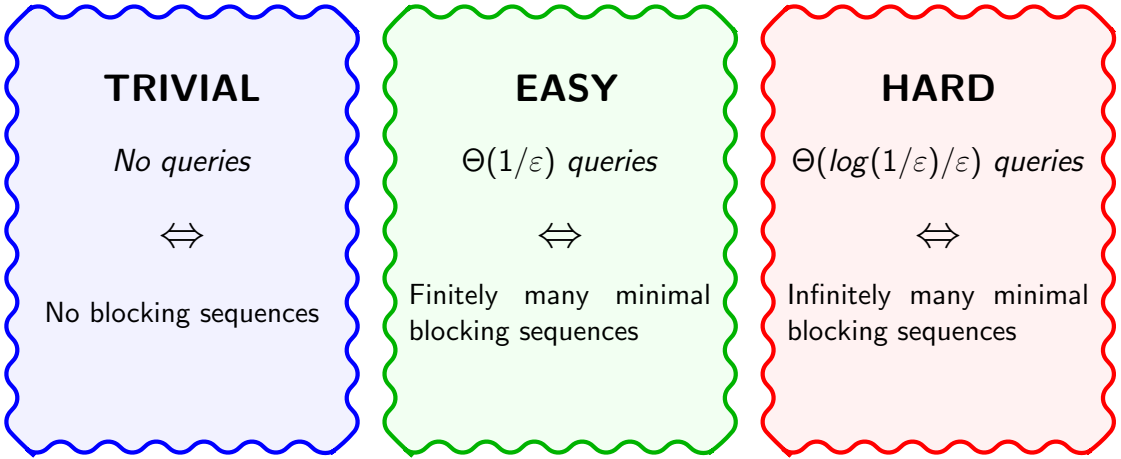

Lower bounds obtained by reduction to the strongly connected case.
If L is given as an NFA, classifying it is **PSPACE-complete**.

Lemma

w is far from $L(\mathcal{A})$ if and only if for all path of SCCs $S_1 \to S_2 \to \cdots \to S_k$ in $\mathcal{A}$, w contains many blocking factors for S_1, then many for S_2,...

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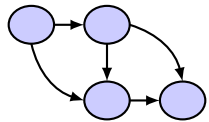

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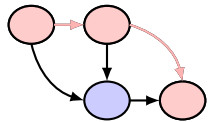

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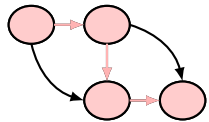

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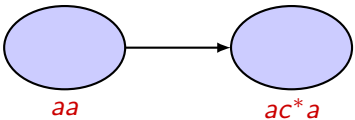

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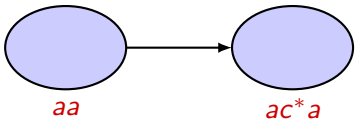

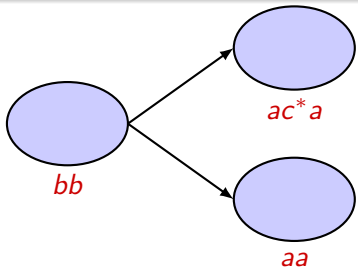

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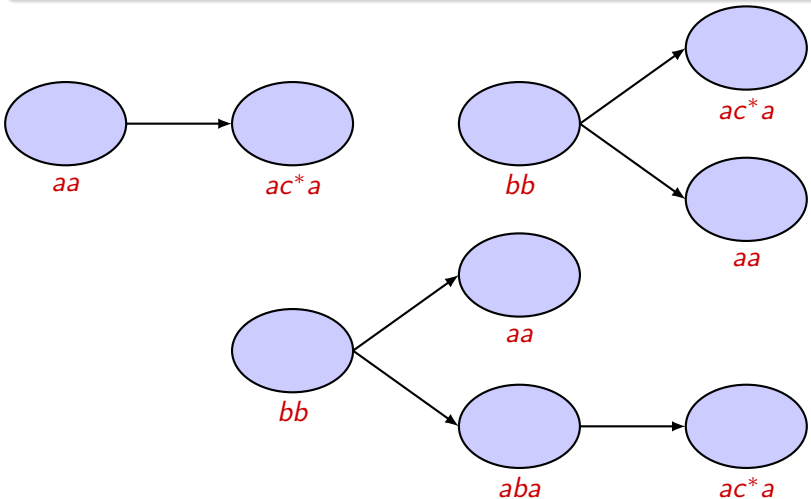

Theorem

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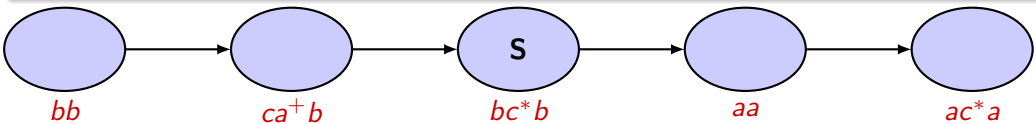

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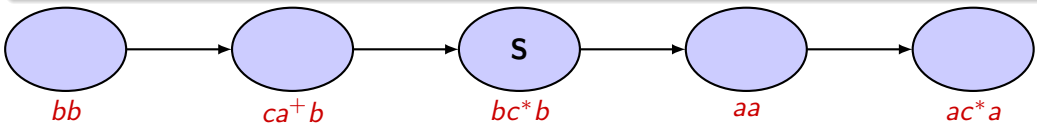

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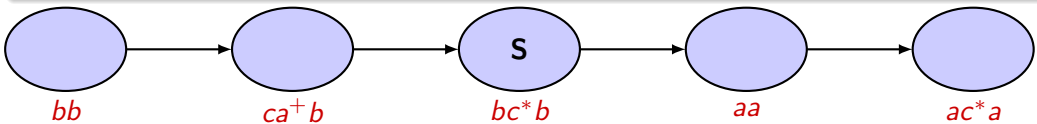

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Testing $L(\mathbf{S})$ is easier than testing $L(\mathcal{A})$.

What is left to do

- Edit distance, complexity for DFAs,...
- Which pushdown languages are testable?
- Tree automata / graphs of bounded treewidth?

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Thank you for your attention!